

Data Science for Innovation

W5: Clustering and Dimensionality Reduction

Presented by
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Overview of Week 5

Today: Clustering and Dimensionality Reduction

Objective

Learn techniques for unsupervised learning, with tools in Python.

Lecture

- Clustering algorithms
- Choosing k
- Evaluating clustering
- Principal Component Analysis
- Eigenvalues and Eigenvectors

Readings

- Intro to Data Mining, Ch. 6
<http://www-users.cs.umn.edu/~kumar/dmbook/ch6.pdf>
- Intro to Data Mining, Ch. 8
<http://www-users.cs.umn.edu/~kumar/dmbook/ch8.pdf>
- Data Science from Scratch, Ch. 11 & 19

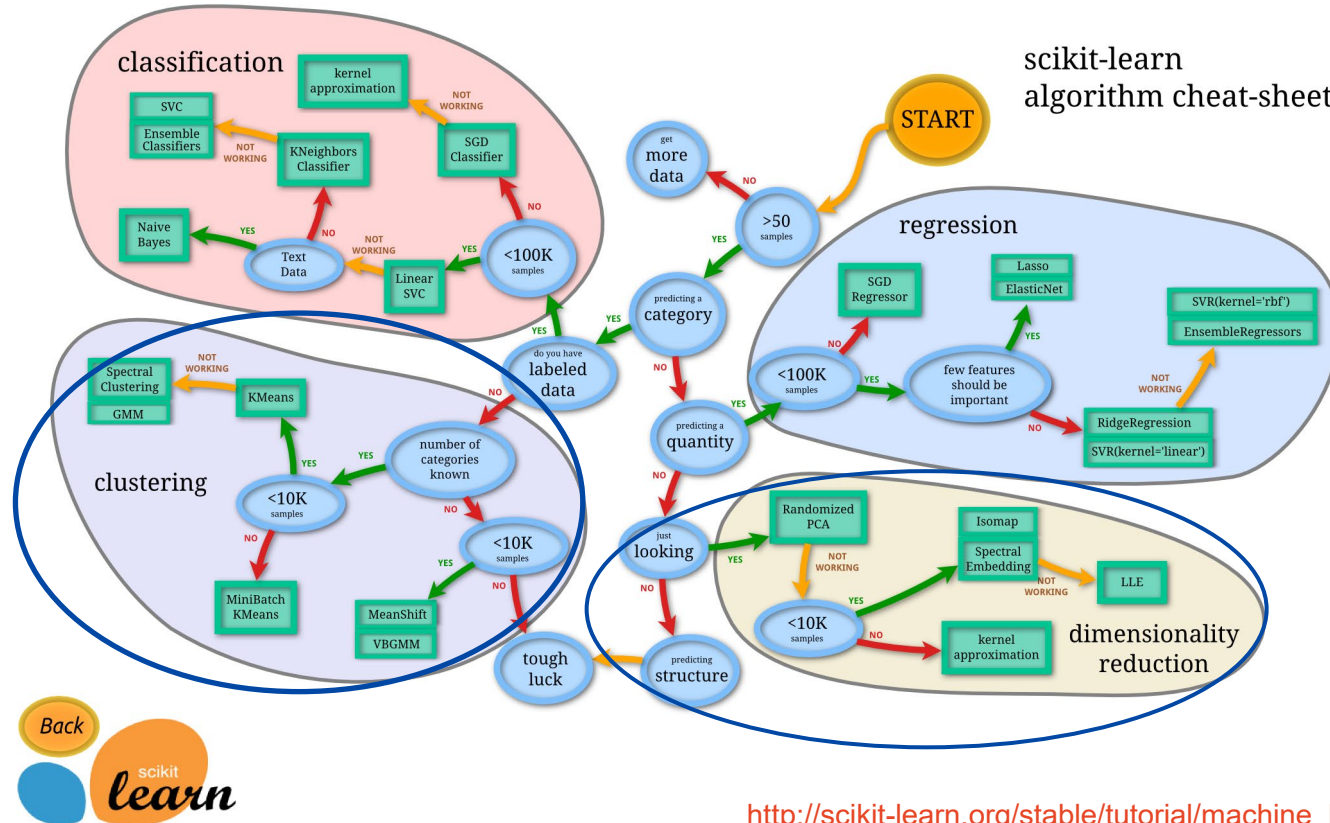
Exercises

- sklearn: clustering and PCA

Unsupervised Learning:

- We'll focus on **unsupervised** machine learning techniques
 - ✓ Association *rule mining*
 - **Dimensionality reduction**
 - *Clustering*
 - Outlier detection
 - Etc.

Machine Learning Map from Scikit-learn



http://scikit-learn.org/stable/tutorial/machine_learning_map/

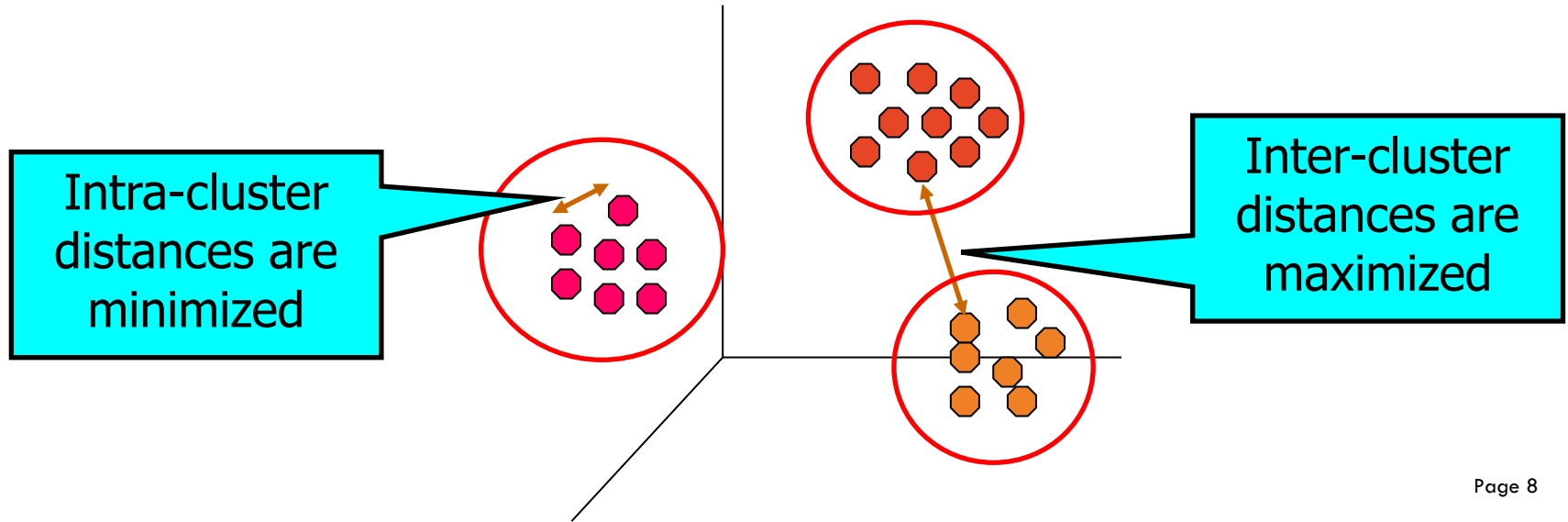
Clustering

Clustering for Understanding

- Group related documents for browsing
- Group genes, proteins, or cells that have similar functionality
- Group stocks with similar price fluctuations
- etc

Clustering: Group Similar Objects

- Group data points into clusters such that
 - Data points in one cluster are more similar to one another.
 - Data points in separate clusters are less similar to one another.
 - Distance function specifies the “closeness” of two objects.



Similarity and Dissimilarity Between Objects

- Distances are normally used to measure the similarity or dissimilarity between two data objects
- Some popular ones include: *Minkowski distance*:

$$d(i, j) = \sqrt[q]{(|x_{i1} - x_{j1}|^q + |x_{i2} - x_{j2}|^q + \dots + |x_{ip} - x_{jp}|^q)}$$

where $i = (x_{i1}, x_{i2}, \dots, x_{ip})$ and $j = (x_{j1}, x_{j2}, \dots, x_{jp})$ are two p -dimensional data objects, and q is a positive integer

- If $q = 1$, d is Manhattan distance

$$d(i, j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}| + \dots + |x_{ip} - x_{jp}|$$

Similarity and Dissimilarity Between Objects (Cont.)

- If $q = 2$, d is Euclidean distance:

$$d(i, j) = \sqrt{(|x_{i1} - x_{j1}|^2 + |x_{i2} - x_{j2}|^2 + \dots + |x_{ip} - x_{jp}|^2)}$$

- Properties

- $d(i, j) \geq 0$
- $d(i, i) = 0$
- $d(i, j) = d(j, i)$
- $d(i, j) \leq d(i, k) + d(k, j)$

Data Structures

- Data matrix
n-observations with p-attributes (measurements).
- Dissimilarity matrix $d(i,j)$ is the dissimilarity between objects i and j
 - expresses the pairwise dissimilarities (distances) between observations in the data set
 - the desired data input to some clustering algorithm

attributes/dimensions

tuples/objects

x_{11}	...	x_{1f}	...	x_{1p}
...	...	...	...	...
x_{i1}	...	x_{if}	...	x_{ip}
...	...	...	...	...
x_{n1}	...	x_{nf}	...	x_{np}

objects

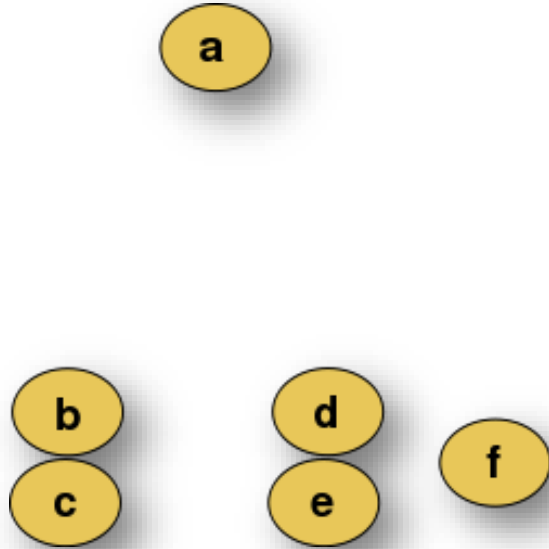
objects

0				
$d(2,1)$	0			
$d(3,1)$	$d(3,2)$	0		
:	:	:		
$d(n,1)$	$d(n,2)$	...	...	0

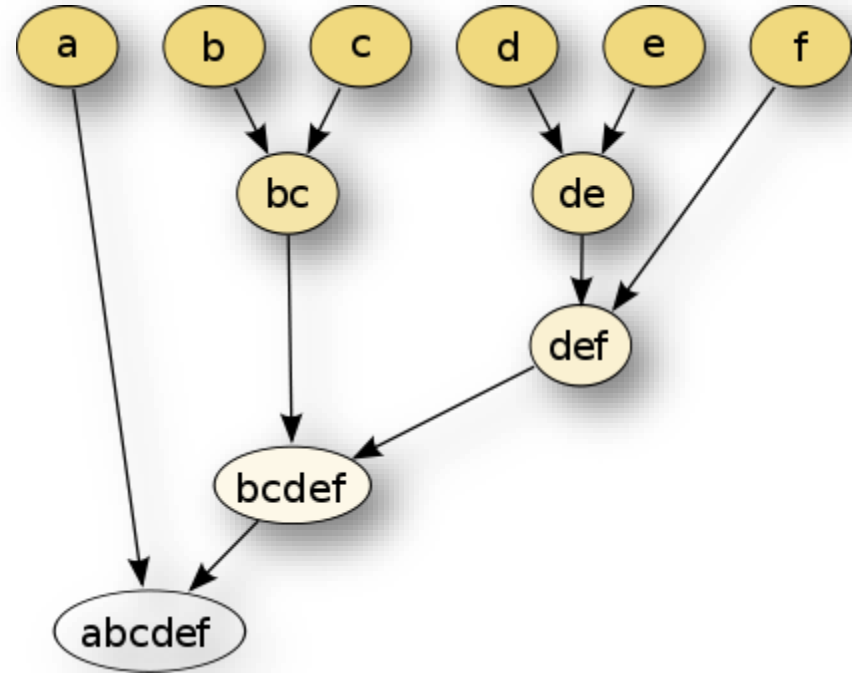
Types of Clusterings

- Important distinction between **hierarchical** and **partitional** sets of clusters
- ***Hierarchical clustering***
A method of cluster analysis which seeks to build a hierarchy of clusters. It produces a set of nested clusters organized as a hierarchical tree
- ***Partitional clustering***
A division data objects into non-overlapping subsets (clusters) such that each data object is in exactly one subset

Hierarchical Clustering



Original Data Items



Hierarchical Data Items

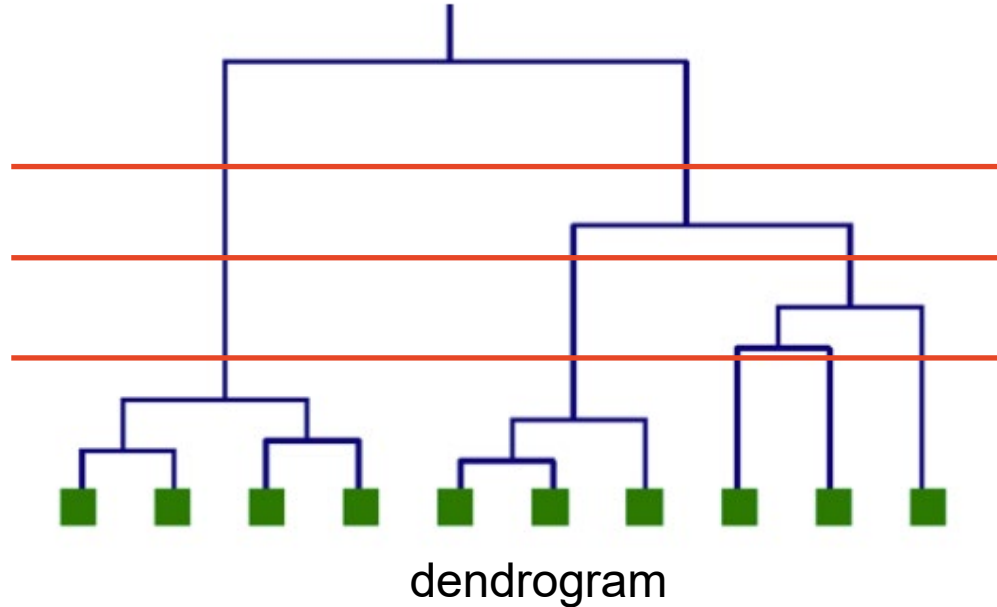
Hierarchical Clustering

Strategies for hierarchical clustering generally fall into two types:

- **Agglomerative:** This is a "bottom up" approach: each object starts in its own cluster, and pairs of clusters are merged as one moves up the hierarchy.
- **Divisive:** This is a "top down" approach: all objects start in one cluster, and splits are performed recursively as one moves down the hierarchy.

Hierarchical Clustering: e.g. Agglomerative

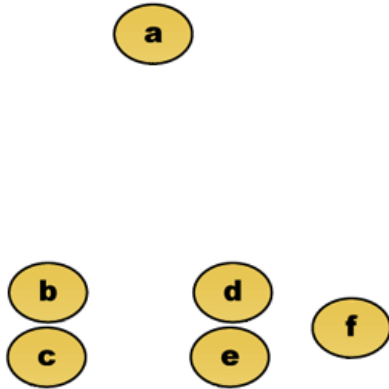
- Initial
 - Each point in its own cluster
- Repeat
 - Find closest pair of clusters
 - Min-distance between any two points
 - Merge them into one cluster
 - Recompute distances between new cluster and others
- Until
 - Desired number of clusters remaining e.g. single cluster



Hierarchical Algorithm

Steps in Hierarchical Algorithm:

- The first step generates the distance calculation matrix for each data item as shown in table below, in this case: {a}, {b}, {c}, {d}, {e}, {f}.



	a	b	c	d	e	f
a	0	184	222	177	216	231
b	184	0	45	123	128	200
c	222	45	0	129	121	203
d	177	123	129	0	46	83
e	216	128	121	46	0	83
f	231	200	203	83	83	0

Hierarchical Algorithm

- Next step is to merge the closest data items.
 - In this case: {b , c} are merged.
 - Therefore, the first clustering process generates: {a}, {b , c}, {d},{e},{f}.

	a	b	c	d	e	f
a	0	184	222	177	216	231
b	184	0	45	123	128	200
c	222	45	0	129	121	203
d	177	123	129	0	46	83
e	216	128	121	46	0	83
f	231	200	203	83	83	0



	a	b,c	d	e	f
a	0	?	177	216	231
b,c	?	0	?	?	?
d	177	?	0	46	83
e	216	?	46	0	83
f	231	?	83	83	0

Hierarchical Algorithm

Distance Calculation between **two hierarchical clusters**:

- single linkage:
 - The minimum distance between elements of each cluster
- complete linkage:
 - The maximum distance between elements of each cluster
- average linkage: i.e. mean distance calculation.

Hierarchical Algorithm with Single Linkage

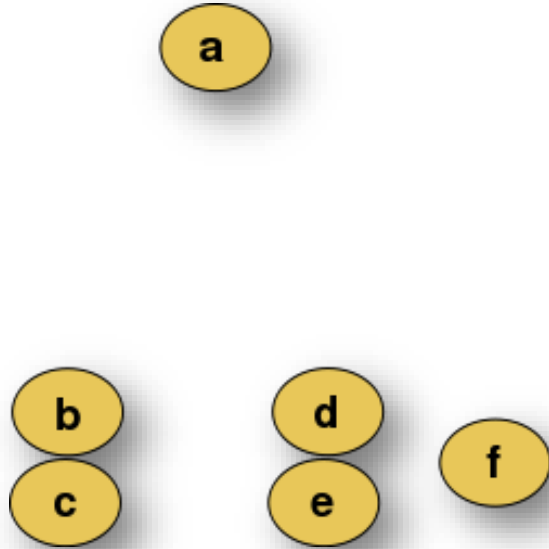
- Repeat the distance calculation process based on single linkage
- Apply merging process based on previous merge results.
 - In this case: {d , e} are merged.
- The final results are: {a}, {b, c} {d, e} → {a}, {b, c}, {d, e, f} → {a}, {b, c, d, e, f} → {a, b, c, d, e, f}

	a	b	c	d	e	f
a	0	184	222	177	216	231
b	184	0	45	123	128	200
c	222	45	0	129	121	203
d	177	123	129	0	46	83
e	216	128	121	46	0	83
f	231	200	203	83	83	0

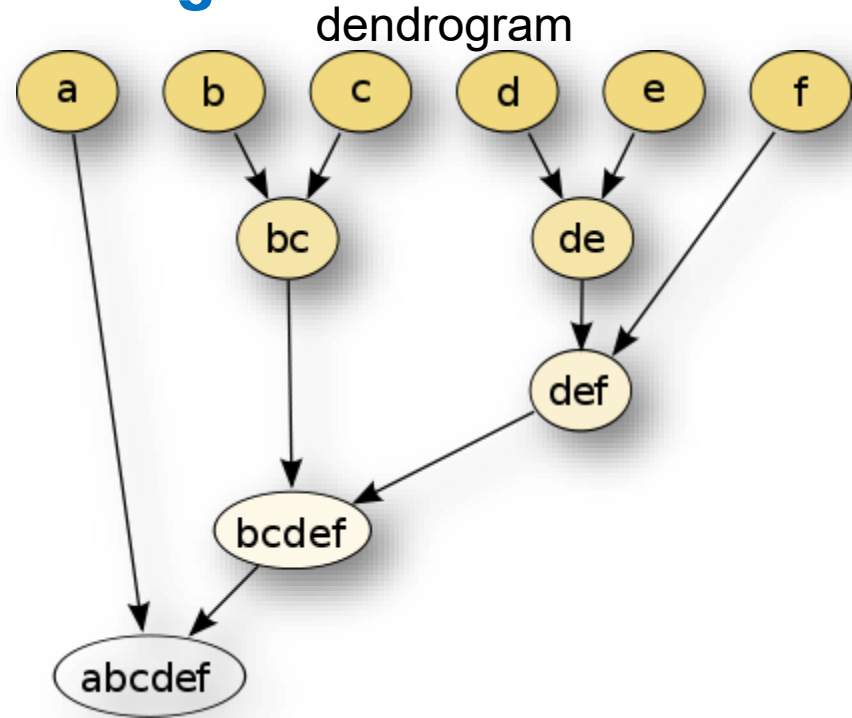


	a	b, c	d	e	f
a	0	184	177	216	231
b, c	184	0	123	121	200
d	177	123	0	46	83
e	216	121	46	0	83
f	231	200	83	83	0

Resultant Hierarchical Clustering

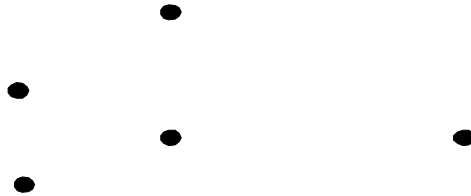
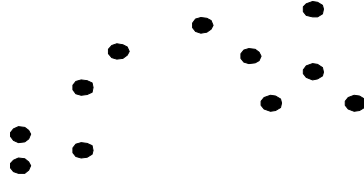


Original Data Items

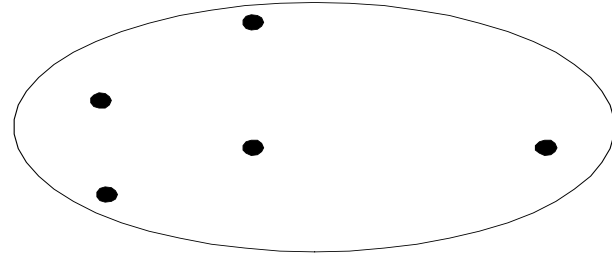
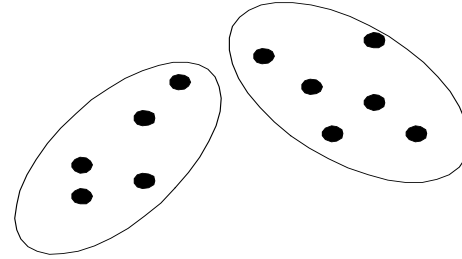


Hierarchical Data Items

Partitional Clustering



Original Points



A Partitional Clustering

Partitional Clustering: K-Means Clustering

- Number of clusters, k , must be specified
- Each cluster is associated with a **centroid** (center point)
- Each point is assigned to the cluster with the closest centroid
- The basic algorithm is very simple

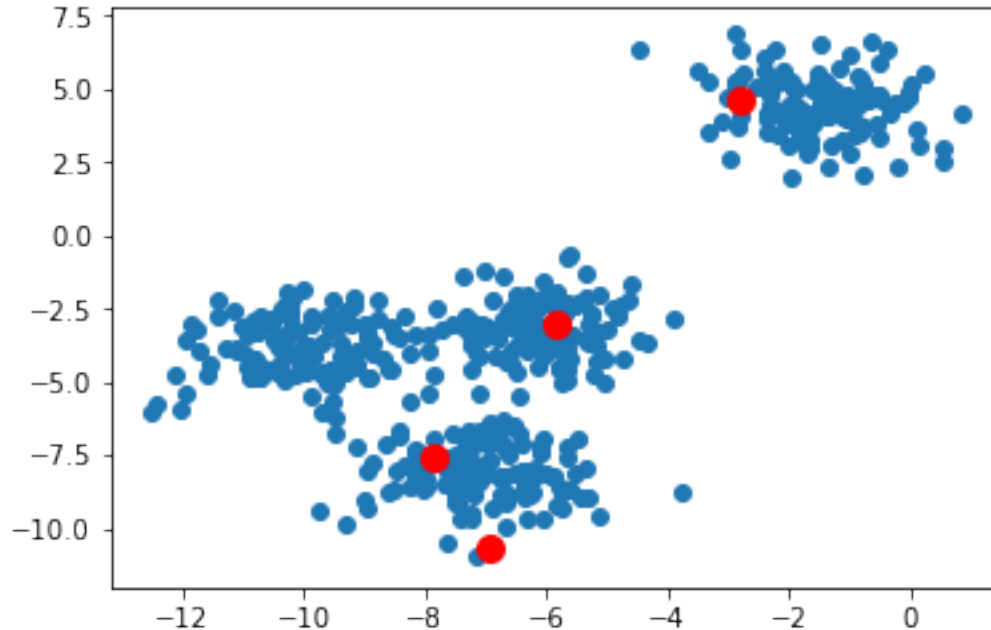
-
- 1: Select K points as the initial centroids.
 - 2: **repeat**
 - 3: Form K clusters by assigning all points to the closest centroid.
 - 4: Recompute the centroid of each cluster.
 - 5: **until** The centroids don't change
-

k-Means Clustering – Details

- Initial centroids are often chosen randomly
- Clusters produced vary from one run to another
- K-means will converge for common similarity measures
- Most of the convergence happens in the first few iterations
- Complexity is $O(n * k * i * d)$
 - n = number of points, k = number of clusters,
 i = number of iterations, d = number of attributes (or dimensions)

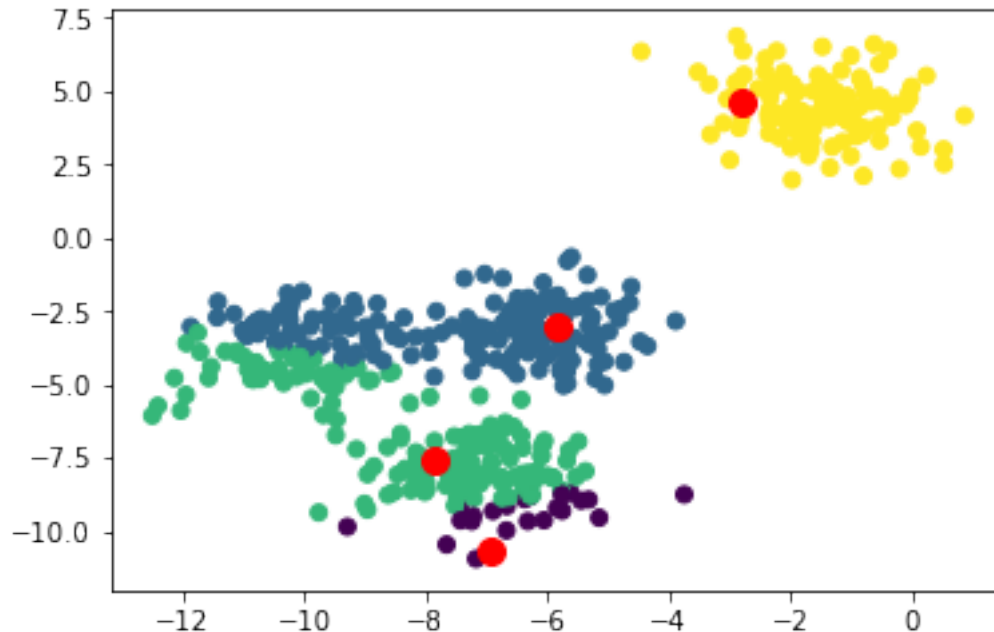
K-Means Example

- Number of clusters: $K = 4$
- Initialize 4 centroids randomly



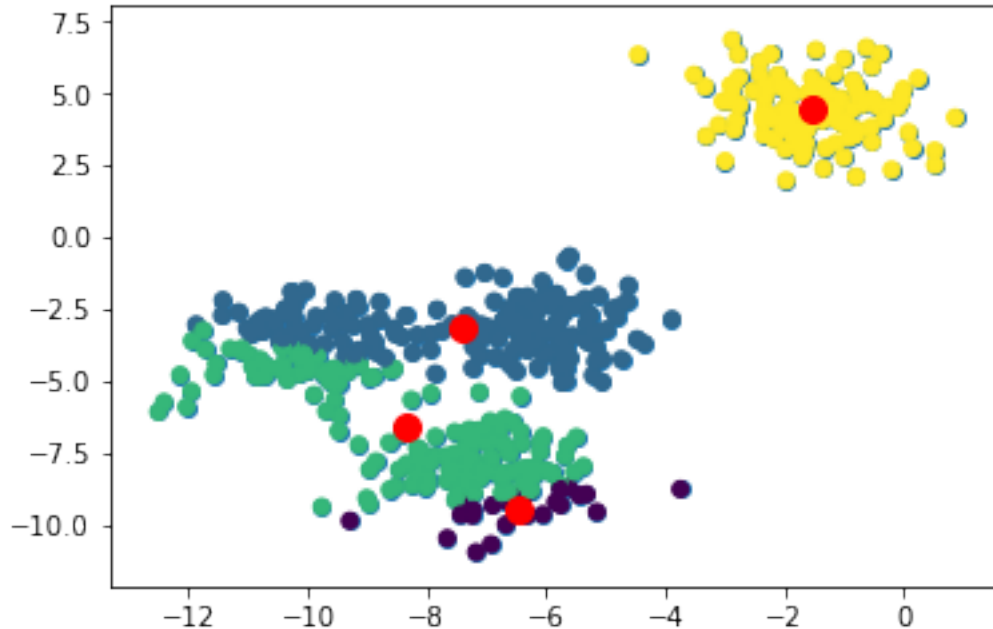
K-Means Example (Cont.)

- Assigned each point to the cluster with the closest centroid



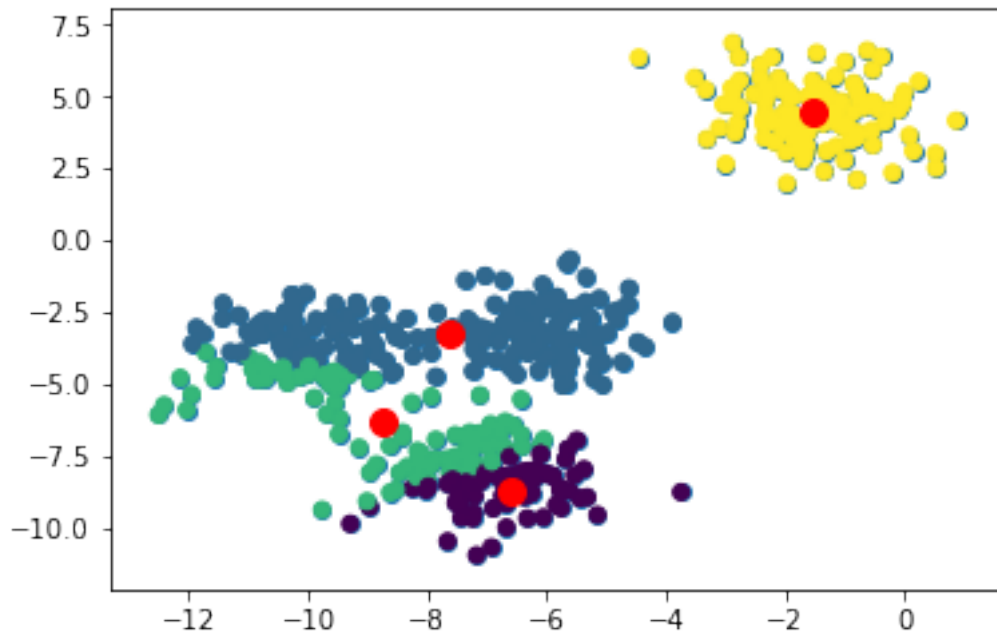
K-Means Example (Cont.)

- Re-compute each cluster center to be the mean of the points previously assigned.



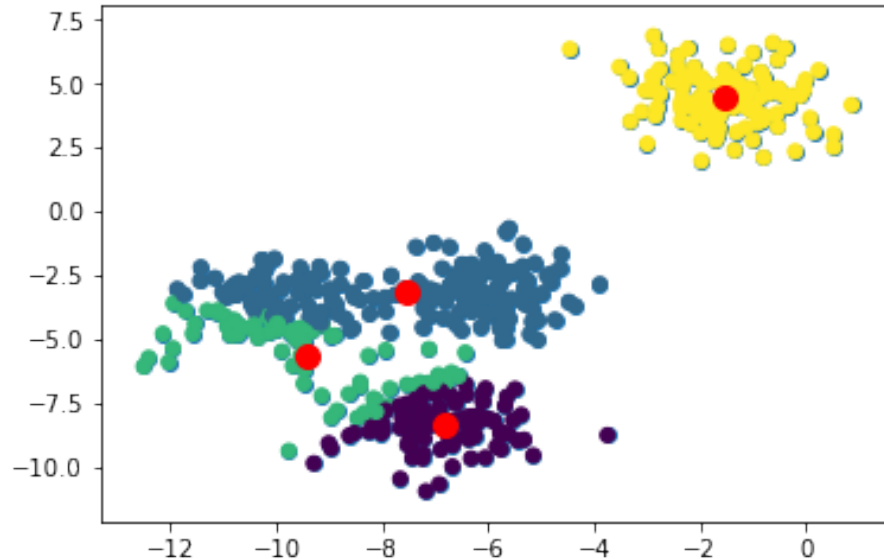
K-Means Example (Cont.)

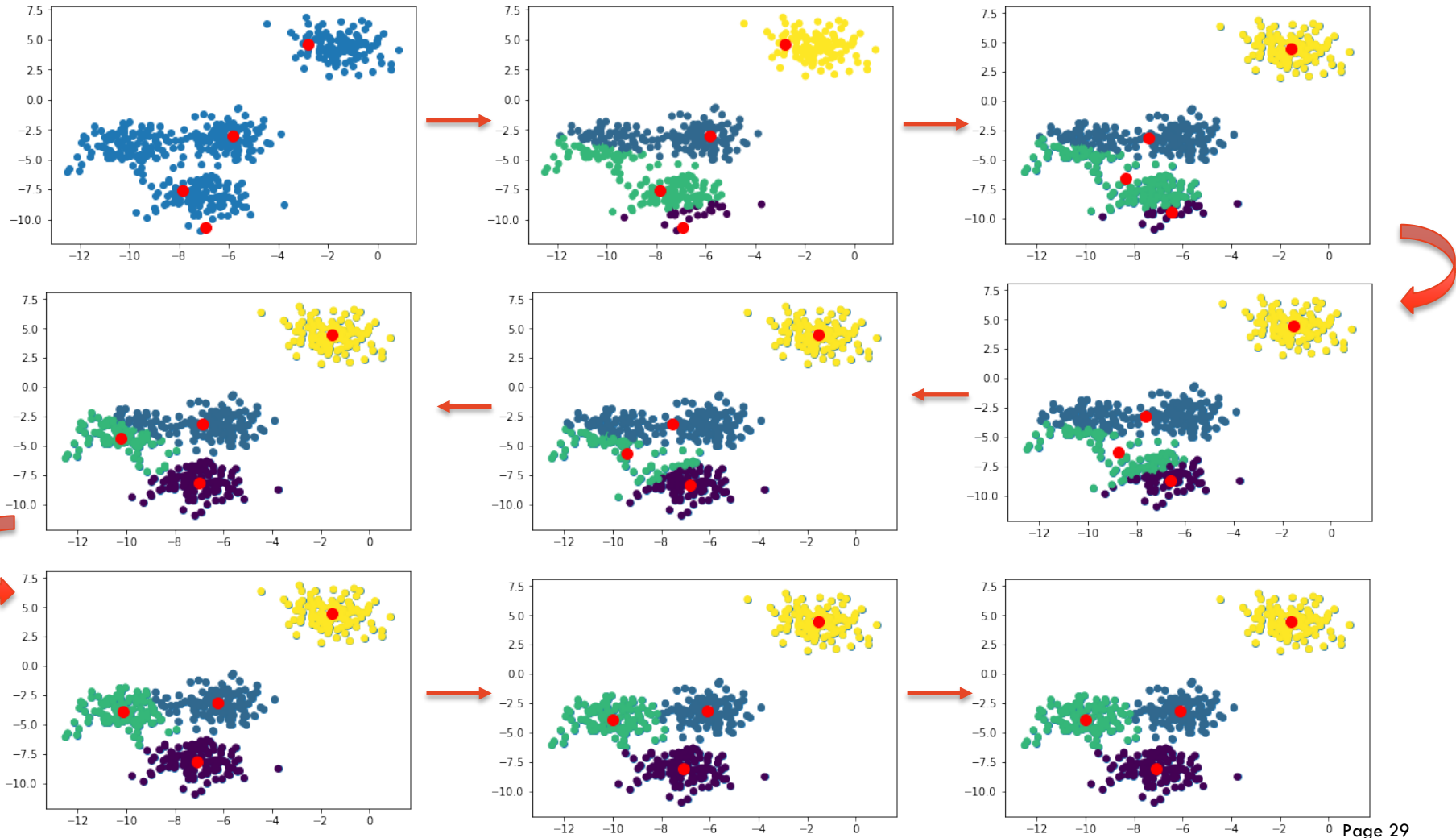
- Assigned each point to the cluster with the closest centroid



K-Means Example (Cont.)

- Re-compute each cluster center
- Centroid changed?
 - Assigned each point to the cluster with the closest centroid
 - Re-compute each cluster center





Exercise: Cluster Analysis

– Clustering

-  code the cells after “Create example data for KMeans clustering from scratch”
-  code cell after “Load handwritten digit dataset”
-  code cell after “Clustering handwritten digits”
- TODO Explore different initialisations

Evaluating Clustering

Cluster Validity

- While clusters are in the eye of the beholder, we may still want to evaluate them...
 - To avoid finding patterns in noise
 - To compare clustering algorithms
 - To compare two sets of clusters
- <http://scikit-learn.org/stable/modules/clustering.html#clustering-evaluation>

Measures of Cluster Validity

- Numerical measures that are applied to judge various aspects of cluster validity, are classified into the following three types.
- **External Index:** Measure the extent to which cluster labels match externally supplied class labels (e.g., accuracy, precision, recall, F1-score)
- **Internal Index:** Measure the goodness of a clustering structure *without* respect to external information (e.g., Sum of Squared Error)
- **Relative Index:** Compare two different clusterings or clusters (often an external or internal index is used)

External Evaluation Measures

- **Homogeneity** ranges from 0 to 1, measuring whether clusters contain data points that are part of a single class
(analogous to precision, $P = TP / (TP + FP)$)
- **Completeness** ranges from 0 to 1, measuring whether classes contain data points that are part of a single cluster
(analogous to recall, $R = TP / (TP + FN)$)
- **V-measure** is the harmonic mean of homogeneity and completeness
(analogous to F1 score = $2PR / (P + R)$)
- <http://scikit-learn.org/stable/modules/clustering.html#homogeneity-completeness-and-v-measure>

Internal: Sum of Squared Error (SSE, or inertia)

- For each point, the error is the distance to the nearest cluster
- To get SSE, we square these errors and sum them.

$$SSE = \sum_{i=1}^K \sum_{x \in C_i} dist^2(m_i, x)$$

- x is a data point in cluster C_i and m_i is the centroid point (mean) for cluster C_i

SSE Example

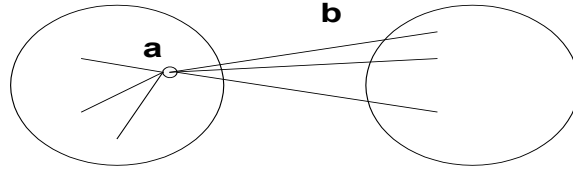
- Suppose we have 3 clusters:
 - Cluster 1: [2, 4] with centroid at 3
 - Cluster 2: [5, 6, 7] with centroid at 6
 - Cluster 3: [8, 10, 12] with centroid at 10
- Squared error for each cluster:
 - $SE1 = (2-3)^2 + (4-3)^2 = 1 + 1 = 2$
 - $SE2 = (5-6)^2 + (7-6)^2 = 1 + 1 = 2$
 - $SE3 = (8-10)^2 + (12-10)^2 = 4 + 4 = 8$
- $SSE = SE1 + SE2 + SE3 = 12$

Internal: Silhouette Coefficient

- For an individual point i
 - Calculate a = average distance of i to points in its cluster
 - Calculate b = average distance of i to points in the next nearest cluster
 - The silhouette coefficient for a point is then given by

$$s = 1 - a/b \quad \text{if } a < b, \quad (\text{or } s = b/a - 1 \quad \text{if } a \geq b, \text{ not the usual case})$$

- The closer to 1 the better



- Silhouette coefficient for dataset is average across all i

Silhouette Coefficient Example

- Suppose we have 3 clusters:
 - Cluster 1 = [[1,0], [1,1]]
 - Cluster 2 = [[1,2], [2,3], [2,2], [1,2]],
 - Cluster 3 = [[3,1], [3,3], [2,1]]
- Take a point [1,0] in cluster 1
- Calculate its average distance to all other points in its cluster, i.e. cluster 1
- So $a_1 = \sqrt{((1-1)^2 + (0-1)^2)} = \sqrt{(0 + 1)} = 1$

Silhouette Coefficient Example (Cont.)

- Now for the point $[1,0]$ in cluster 1 calculate its average distance from all the objects in cluster 2 and cluster 3.
- Of these take the minimum average distance.
- So for cluster 2:
 - $[1,0] \rightarrow [1,2] = \text{distance} = \sqrt{((1-1)^2 + (0-2)^2)} = \sqrt{(0+4)} = 2$
 - $[1,0] \rightarrow [2,3] = \text{distance} = \sqrt{((1-2)^2 + (0-3)^2)} = \sqrt{(1+9)} = 3.16$
 - $[1,0] \rightarrow [2,2] = \text{distance} = \sqrt{((1-2)^2 + (0-2)^2)} = \sqrt{(1+4)} = 2.24$
 - $[1,0] \rightarrow [1,2] = \text{distance} = \sqrt{((1-1)^2 + (0-2)^2)} = \sqrt{(0+4)} = 2$
- Therefore, the average distance of point $[1,0]$ in cluster 1 to all the points in cluster 2 =

$$(2+3.16+2.24+2)/4 = 2.35$$

Silhouette Coefficient Example (Cont.)

– Similarly, for cluster 3.

– $[1,0] \rightarrow [3,1] = \text{distance} = \sqrt{((1-3)^2 + (0-1)^2)} = \sqrt{(4+1)} = 2.24$

– $[1,0] \rightarrow [3,3] = \text{distance} = \sqrt{((1-3)^2 + (0-3)^2)} = \sqrt{(4+9)} = 3.61$

– $[1,0] \rightarrow [2,1] = \text{distance} = \sqrt{((1-2)^2 + (0-1)^2)} = \sqrt{(1+1)} = 1.41$

– Therefore, the average distance of point $[1,0]$ in cluster 1 to all the points in cluster 3 =

$$(2.24+3.61+1.41)/3 = 2.42$$

– Now, the minimum average distance of the point $[1,0]$ in cluster 1 to the other clusters 2 and 3 is,

$$b1 = 2.35 \quad (2.35 < 2.42)$$

Silhouette Coefficient Example (Cont.)

- So the silhouette coefficient of point $[1,0]$ in cluster 1

$$s_1 = 1 - (a_1/b_1) = 1 - (1/2.35) = 1 - 0.43 = 0.57$$

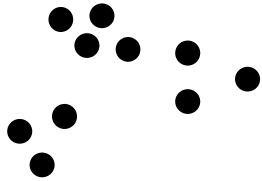
- In a similar fashion you need to calculate the silhouette coefficient for each data point in each cluster
- Then we average them to calculate the overall silhouette coefficient to evaluate the resultant clusters
- The closer to 1 the better

Exercise: Evaluation

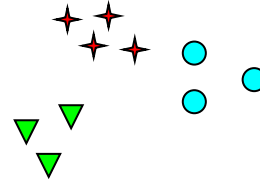
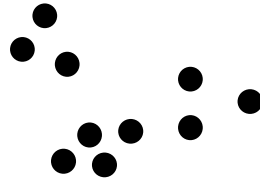
- Evaluating with respect to a gold partition
 -  code cell after “Evaluating clustering”
 -  code cell after “Comparing initialisations”
 - TODO Discuss evaluation output

Choosing k

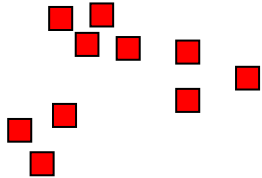
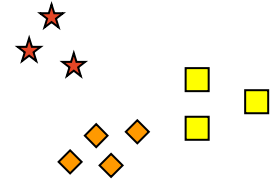
Notion of a Cluster can be Ambiguous



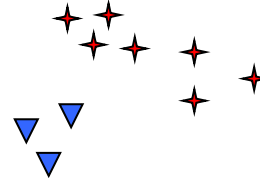
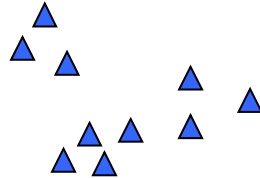
How many clusters..?



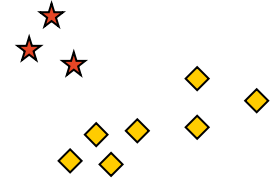
Maybe six clusters



Two clusters?



Or four clusters

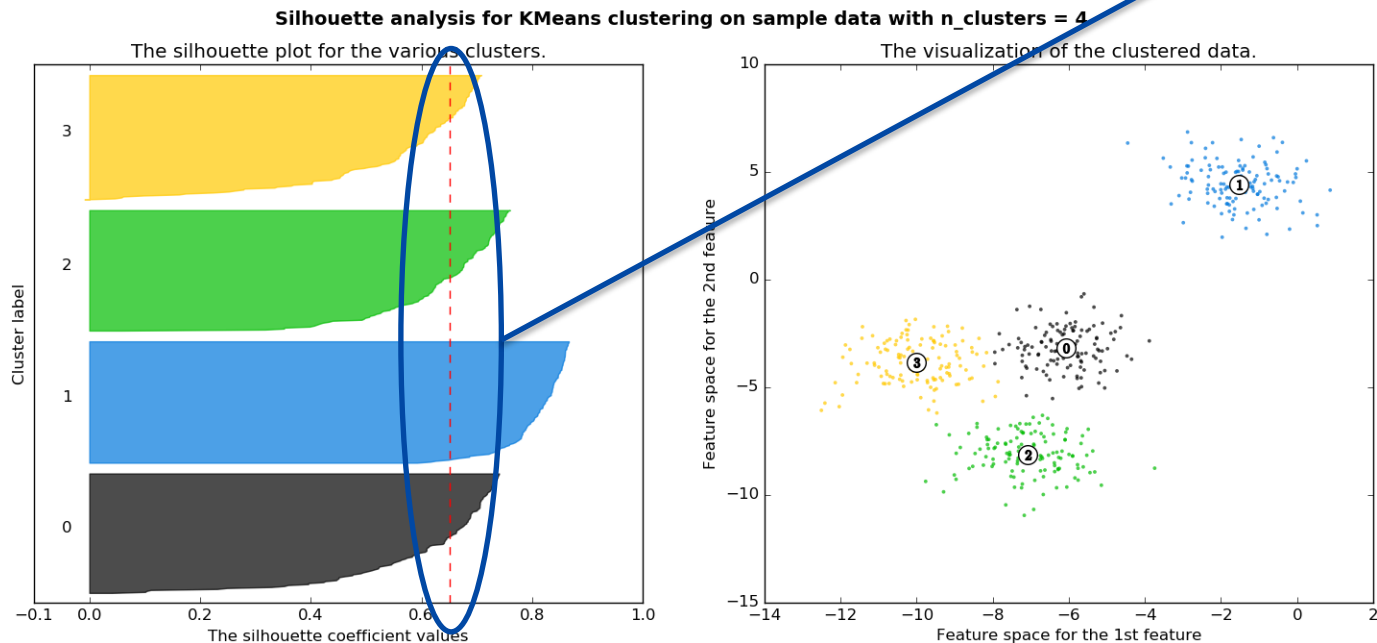


Choosing k

- Often we don't know the number of clusters in our data
- Selecting k is generally an interactive process
- There are some approaches to aid interactive selection, and sometimes automate it

Using Silhouettes to choose k

High average silhouette indicates points far away from neighbouring clusters

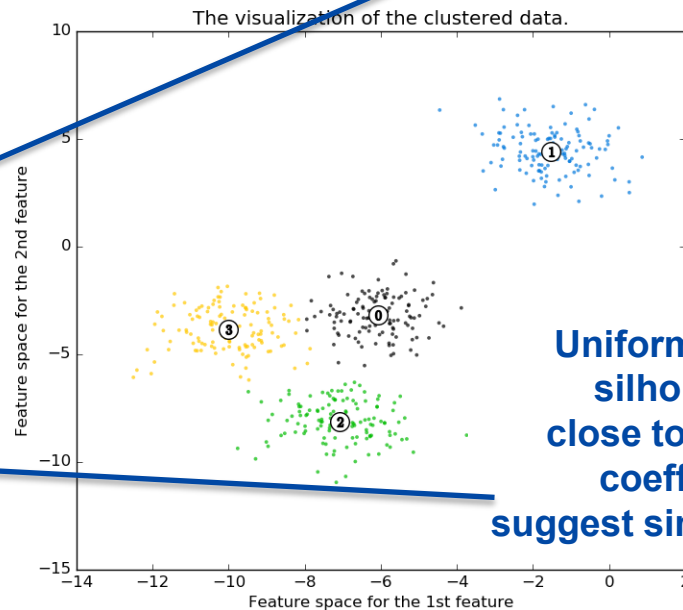
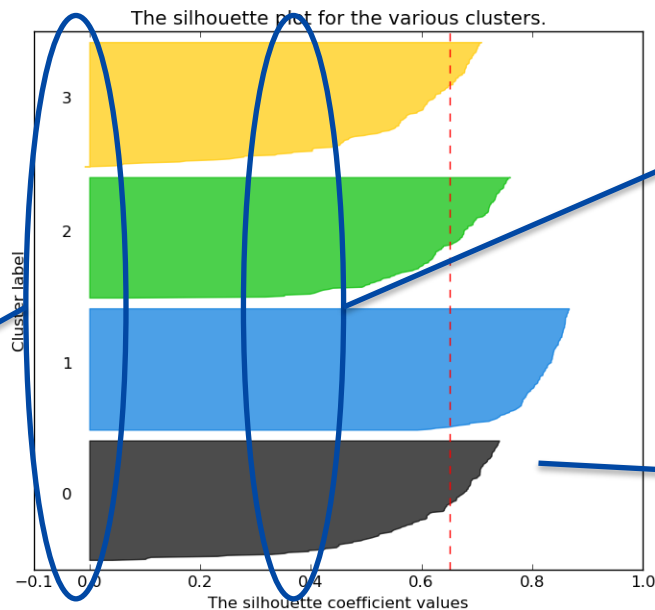


http://scikit-learn.org/stable/auto_examples/cluster/plot_kmeans_silhouette_analysis.html

Using Silhouettes to choose k

Bar chart showing silhouette values for each item grouped by cluster

Silhouette analysis for KMeans clustering on sample data with $n_clusters = 4$



http://scikit-learn.org/stable/auto_examples/cluster/plot_kmeans_silhouette_analysis.html

Exercise: Choosing k

- Selecting the number of clusters
 -  code cell after “Create example data for choosing k ”
 -  code cell after “Choosing k using silhouette analysis”
 - TODO Choosing k by plotting sum of squared error

Pre-Processing for Clustering

- We only looked at choosing k , but more pre-processing needed
- Data Cleansing
- Data Transformation
- **Data Normalisation**
- **Dimensionality Reduction** / choice or projection of dimensions
 - closely related to choice of **distance metric**
 - we used Euclidean Metric so far (L_2 -Norm)
 - but others possible too, eg. Manhattan Distance (L_1 -Norm)
 - "Curse of high dimensionality"; cf Aggarwal et al.: "On the Surprising Behavior of Distance Metrics in High Dimensional Space", 2001.

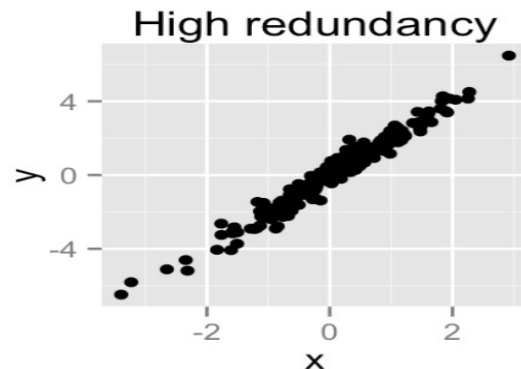
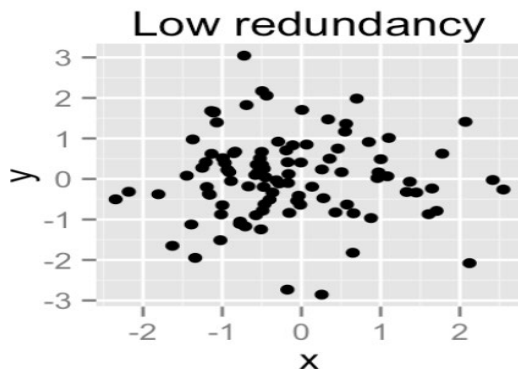
Principal Component Analysis

Principal Components Analysis

- It aims transforming the original data from high dimensional space into lower dimensional space.
- The new variables in the lower dimensional space corresponds to a linear combination of the originals and are called **principal components (PC)**
- PCA helps in
 - **Visualization.** Using the right variables to plot items will give more insights.
 - **Uncovering Clusters.** With good visualizations, hidden categories or clusters could be identified.
 - **Dimensionality reduction.** Reduce number of dimensions in data

Principal Components Analysis

- PCA method is particularly useful when the variables within the data set are **highly correlated**.
- **Correlation** indicates that there is redundancy in the data.
- Correlation is captured by the covariance matrix¹.
- PCA is traditionally performed on **covariance** matrix or correlation matrix.



¹covariance matrix is a square **matrix** that contains the variances and **covariances** associated with several variables.

Covariance Matrix

- Representing Covariance between dimensions as a matrix e.g for three attributes (x,y,z):

$$C = \begin{pmatrix} \text{cov}(x,x) & \text{cov}(x,y) & \text{cov}(x,z) \\ \text{cov}(y,x) & \text{cov}(y,y) & \text{cov}(y,z) \\ \text{cov}(z,x) & \text{cov}(z,y) & \text{cov}(z,z) \end{pmatrix}$$

Variances

- The covariance between one dimension and itself is the variance
 - Diagonal is the variances of x, y and z
- $\text{cov}(x,y) = \text{cov}(y,x)$ hence matrix is symmetrical about the diagonal
- N-dimensional data will result in NxN covariance matrix

Covariance Matrix Example

- Below is the covariance matrix of some 3 variables.
- Their variances are on the diagonal, and the sum of the 3 values (3.448) is the overall variability

1.343730	-.1601522	.1864702
-.1601522	.61920562	-.1266842
.1864702	-.1266842	1.485549

- The diagonal elements are the **variances** of the different variables.
- In the covariance table above, the off-diagonal values are different from zero. This indicates the presence of redundancy in the data.
- In other words, there is a certain amount of correlation between variables.

PCA Example

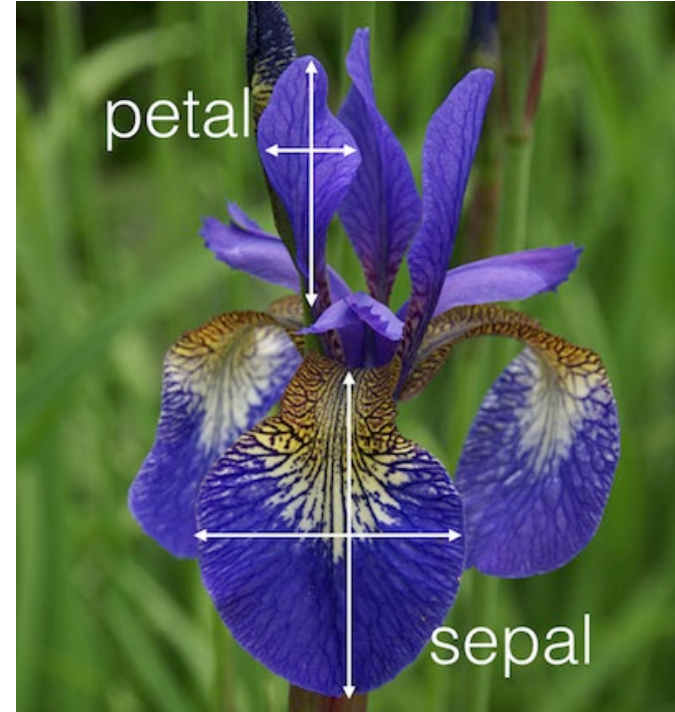
- PCA creates uncorrelated PC variables (called eigenvectors) having zero covariations and variances (called eigenvalues) sorted in decreasing order.
- The first PC captures the greatest variance , the second greatest variance is the second PC, and so on.
- By eliminating the later PCs we can achieve dimensionality reduction.
 - The 1st PC accounts for or "explains" $1.651/3.448 = 47.9\%$ of the overall variability;
 - the 2nd one explains 35.4% of it; the 3rd one explains 16.7% of it.

1.65135	.000000	.000000
.000000	1.220288	.000000
.000000	.000000	.576843

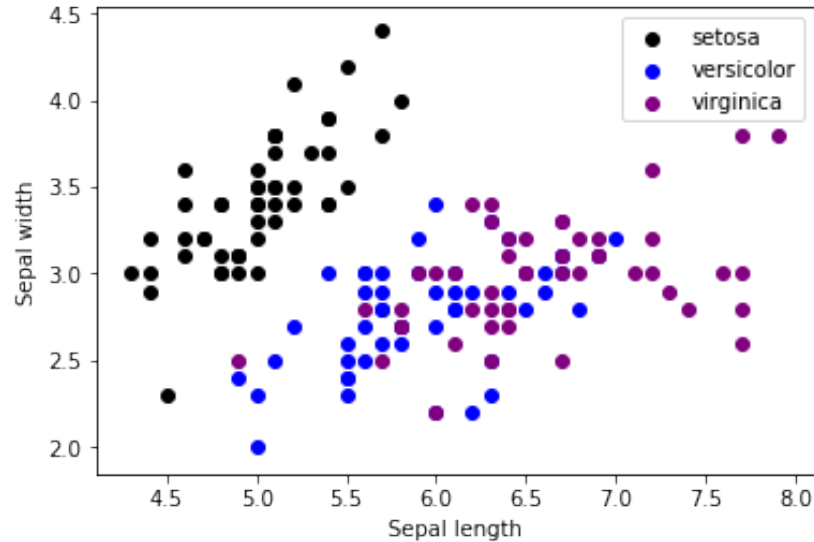
The covariance matrix
between the principal
components

PCA on Iris Dataset

- Iris data has 150 observations equally distributed among three species:
Setosa, Versicolor and Verginica.
- It has four variables:
 - Sepal length and width
 - Petal length and width
- Which variables I can use to plot the data in two dimensional space?
- Lets try using the two features:
Sepal length VS. Sepal width

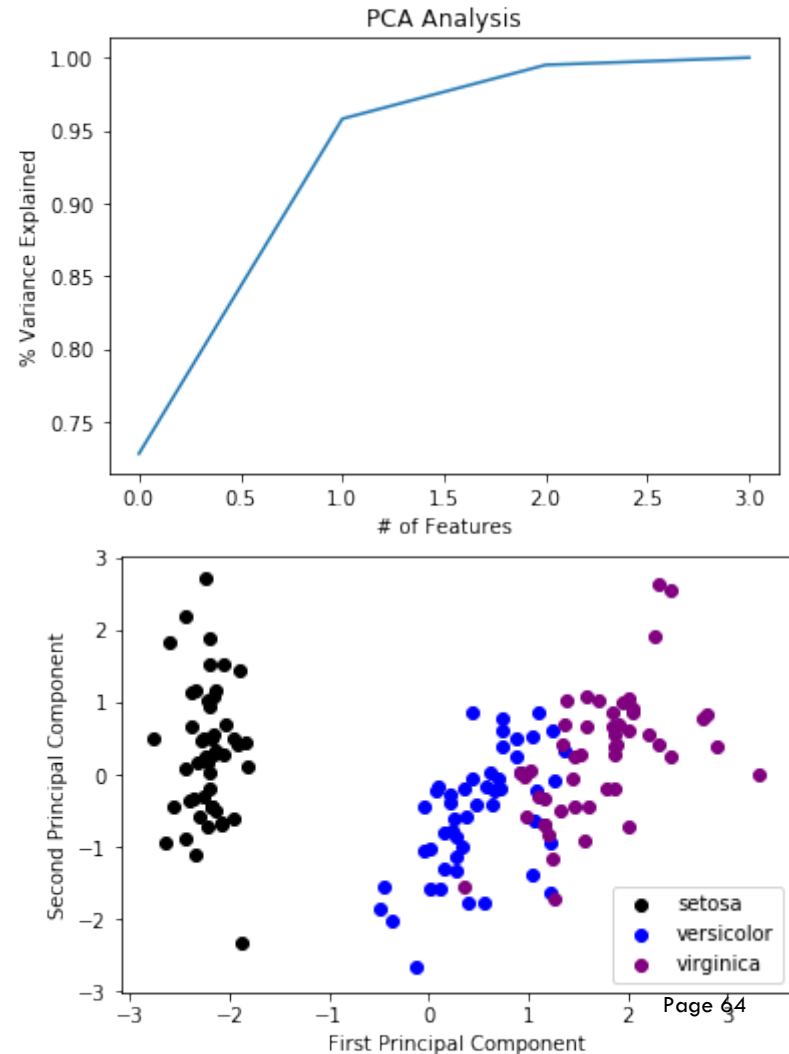


Plotting the data points using Sepal Length vs Sepal Width



PCA on IRIS Dataset

- Lets now choose the best variables using PCA and then plot the data
- The eigenvalues are:
[0.728 0.230 0.037 0.005]
- The first two PCs represent 95.8% of the variance of the data
- Which means we can reduce the data into two dimensional spaces by eliminating PC3 and PC4



Exercise: Dimensionality Reduction

- Selecting the number of clusters
 -  code cell after “Dimensionality Reduction”
 -  code cell after “Deciding how many componenets”
 - TODO PCA on digits dataset

Review

Other tools and Techniques (not examinable)

- Scikit-learn user guide, § 4.4 (Dimensionality reduction).
http://scikit-learn.org/stable/modules/unsupervised_reduction.html
- Scikit-learn user guide, § 2.7 (Outlier detection).
http://scikit-learn.org/stable/modules/outlier_detection.html
- Etc

Assignment 2: Project Stage 2

Project Stage 2: Experiment, Quantify, Report

Objective

Complete a piece of data science work to answer a question or provide an intelligent data-driven tool.

Activities

- Define experimental framework
- Perform analysis or build tool
- Describe evaluation and conclusions

Output

- ~4-page report describing framework, analysis and conclusions (plus code)
- Demonstration (2-3/3-4 mins)

Marking

- 40% of overall mark
 - 30% report and code
 - 10% presentation

Suggested timeline for Assignment 2 (project stage 2)

- *W7: Define experimental framework*
- W8: Implement approach
- W9: Write first page (framework, approach)
- W10: Evaluate and benchmark approach
- W11: Analyse and characterise results
- W12: Submit full report (W9 + results, analysis, conclusions)
- W12: Deliver demonstration